Systemadministration

Linux

Från UNIX till Linux

Agenda

1. Kort repetition
2. Historia
3. Licenser
4. Används *nix idag?
5. Vad är Linux
   och hur installerar man det?
6. Vad kommer vi använda

Kort repetition

Source Code & Machine Code

Källkod & maskinkod

```c
static void
next_line_num (void)
{
  char *endp = line_num_end;
  do
    {
      if ((*endp)++ < '9')
        return;
      *endp-- = '0';
    }
  while (endp >= line_num_start);

  if (line_num_start > line_buf)
    *--line_num_start = '1';
  else
    *line_buf = '>';
  if (line_num_start < line_num_print)
```

Kompileras

```
03 00 3e 00 01 00 00 00  f0 31 00 00 00 00 00 00
40 00 00 00 00 00 00 00  18 a2 00 00 00 00 00 00
00 00 00 00 40 00 38 00  0d 00 40 00 1e 00 1d 00
06 00 00 00 04 00 00 00  40 00 00 00 00 00 00 00
40 00 00 00 00 00 00 00  40 00 00 00 00 00 00 00
d8 02 00 00 00 00 00 00  d8 02 00 00 00 00 00 00
08 00 00 00 00 00 00 00  03 00 00 00 04 00 00 00
18 03 00 00 00 00 00 00  18 03 00 00 00 00 00 00
18 03 00 00 00 00 00 00  1c 00 00 00 00 00 00 00
1c 00 00 00 00 00 00 00  01 00 00 00 00 00 00 00
01 00 00 00 04 00 00 00  00 00 00 00 00 00 00 00
```

GNU Coreutils contributors. (2023, July 31). cat.c. In *GNU Savannah*. Retrieved 07:25, August 1, 2023, from https://git.savannah.gnu.org/cgit/coreutils.git/tree/src/cat.c

Source Code

Machine Code

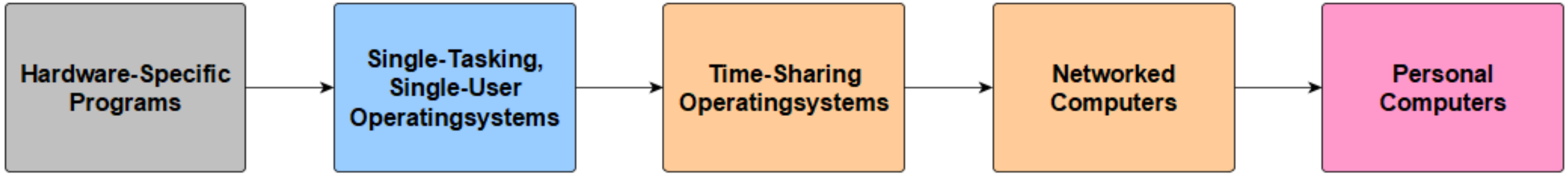
Operating System

Består utav:

- **Kernel/kärna**
  Ett program som alltid körs
  ‘Är operativsystemet’
- **Systemprogram**
  Program som inte är en del av kärnan
- **Användarprogram**
  Övriga program, driftar inte systemet

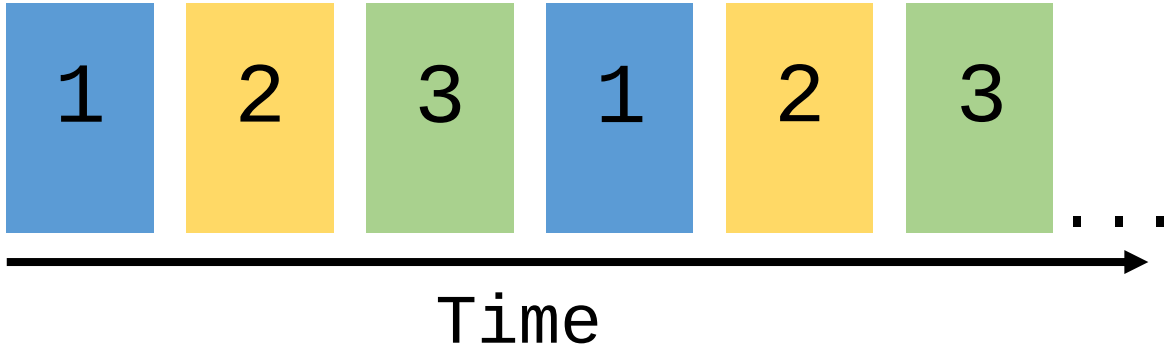
Time Sharing

En dator flera användare & flera processer samtidigt!

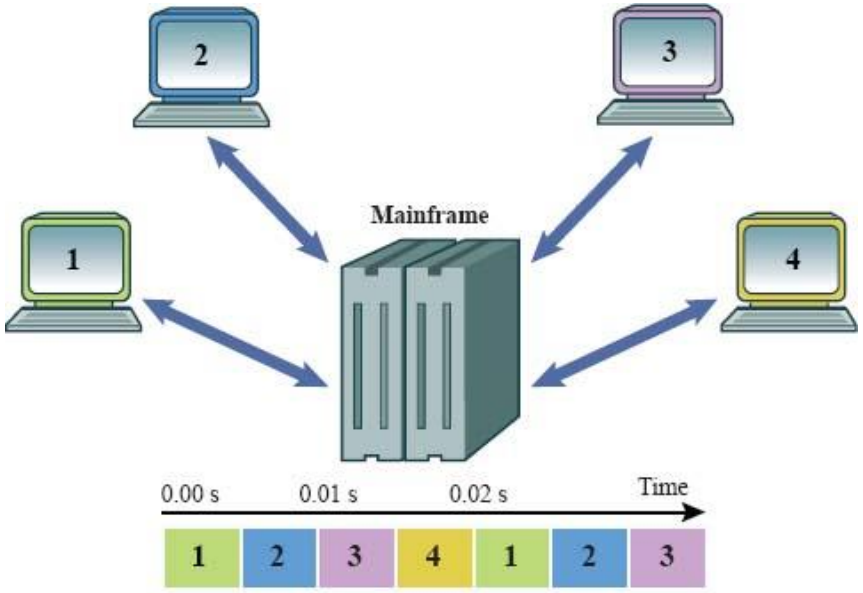


Time Sharing

Multi-Tasking

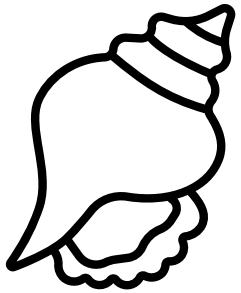


Multi-User



Computing by user over time

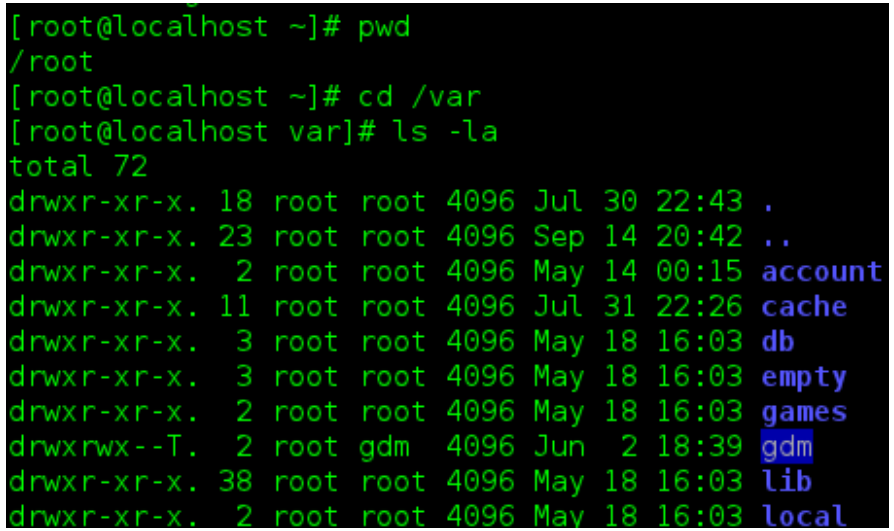
Shell & User Interface



GUI **G**raphical
User
Interface

CLI **C**ommand-**L**ine
Interface

Microsoft Windows 10 Home, version 22H2, by Microsoft Coporation, 2022, Wikimedia Commons (https://en.wikipedia.org/w/index.php?curid=71854645). Fair use/I enlighet med citaträtt, trademark of Microsoft Corporation.

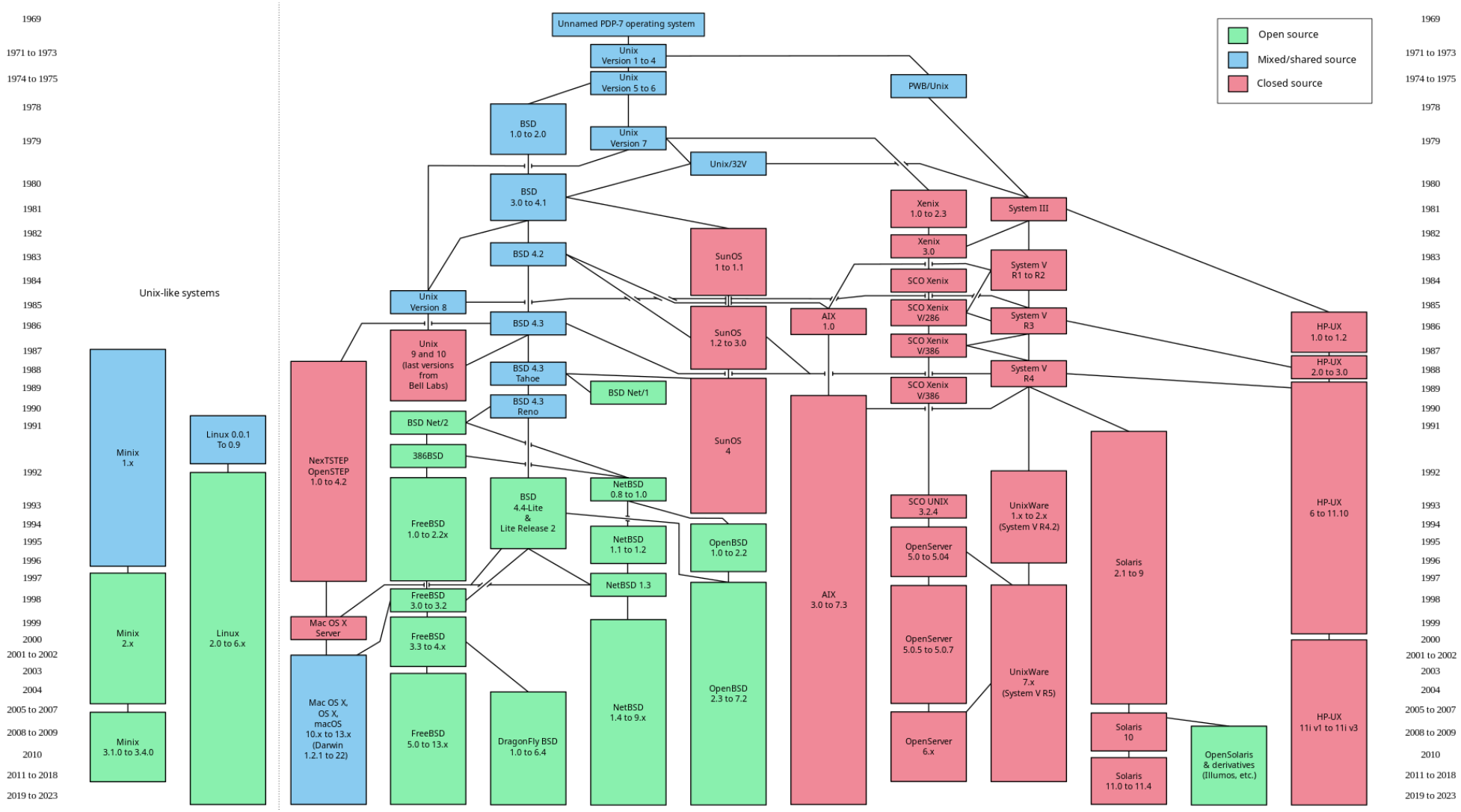


```
[root@localhost ~]# pwd
/root
[root@localhost ~]# cd /var
[root@localhost var]# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x.  2 root root 4096 May 14 00:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:26 cache
drwxr-xr-x.  3 root root 4096 May 18 16:03 db
drwxr-xr-x.  3 root root 4096 May 18 16:03 empty
drwxr-xr-x.  2 root root 4096 May 18 16:03 games
drwxrwx--T.  2 root gdm  4096 Jun  2 18:39 gdm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x.  2 root root 4096 May 18 16:03 local
```

Sample of Bash in GNOME Terminal. Screenshot taken in Fedora. Results of ping, pwd, cd, yum and ls command., by ZxxZxxZ, 2011, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=16485488). No license.

Historia

UNIX History …and Linux History



A diagram showing the key Unix and Unix-like operating systems, by Eraserhead1, Infinity0, Sav_vas, 2008, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Unix_history-simple.svg). CC BY-SA 3.0

Multics [1964]

Ett tidigt **time-sharing** OS

Inledningsvis ett samarbete mellan

- MIT
- General Electric
- **Bell Labs**

Vectorized Logo of Multics, by VectorVoyager, 2022, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Multics-logo.svg). CC BY-SA 4.0 International.

Multics [1964]

Utvecklades på GE-645 mainframe
kostade då $10 miljoner

GE-645 System, by Fernando Corbató, 2012, Multicians (https://www.multicians.org/645artist.html). Fair use/I enlighet med citaträtt

Bell Labs

The logo of Bell laboratories from 1969 to 1983, by Bell Laboratories, 1969, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Bell_Laboratories_logo.svg). Public domain, trademark of Nokia

Anrikt forskningsföretag inom teknikutveckling

B.la. 1956 Nobelpriset i fysik för halvledaren

In 1997 Lucent Technologies created this replica to commemorate the 50th anniversary of the invention of the point-contact transistor at Bell Labs in December 1947, by Federal employee, 1997, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Replica-of-first-transistor.jpg). Public domain

AT&T
American Telephone & Telegraph Company

1925-1996 **AT&T Bell Labs**

1877-1982 tillsatt monopol på telekommunikation

ingen verksamhet utanför sitt mandat

Logo of the AT&T Corporation during the break-up of the Bell System in 1983, designed by Saul Bass. 12-bar variant used from 1983 to 2003, vertical lockup, by AT&T Coproration, circa 1983, Wikipedia (https://en.wikipedia.org/wiki/File:AT%26T_logo_12-bar_vertical_lockup.svg). Public domain, trademark of AT&T Corporation

UNIX [1969]

Ken Thompson & **Dennies Ritchie** m.fl. på **Bell Labs** utvecklade…

- programmeringsspråket **C**
- **Unics** (senare **UNIX**) utifrån sina lärdomar i **Multics**

UNIX skulle vara simpelt

UNIX philosophy

Ken Thompson

Dennis Ritchie

[1]

[2]

PDP-11 kostade då $11,000

Research UNIX [1971-1989]

UNIX fick dock inte säljas…

Gavs ut tillsammans med sin **källkod**

AT&T
American Telephone & Telegraph Company

1877-1982 tillsatt monopol på telekommunikation

ingen verksamhet utanför sitt mandat

Research UNIX Derivatives

Flera företag, produkter & forskning uppkom runt **UNIX**, b.la.

BSD wordmark, by The FreeBSD Project, circa 2009, Wikipedia (https://en.wikipedia.org/wiki/File:BSD_wordmark.svg). Public domain, trademark of Computer Systems Research Group

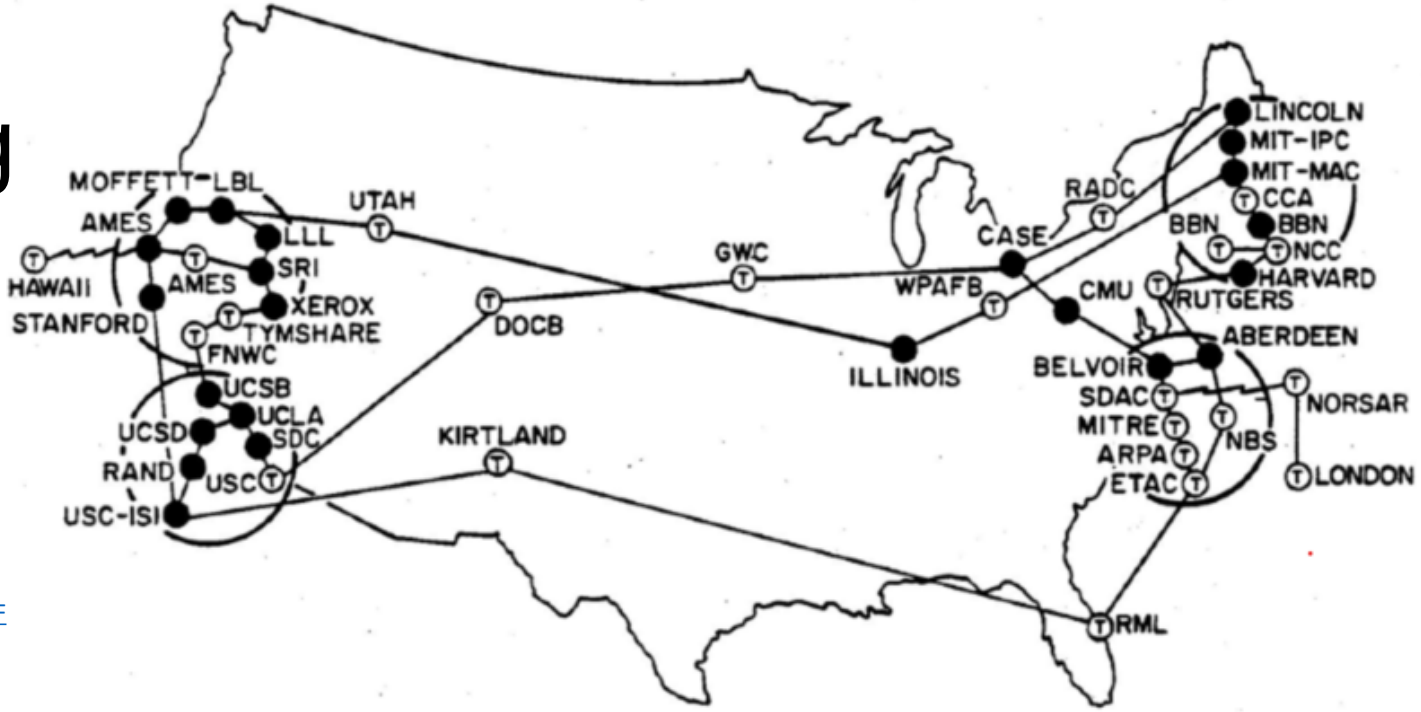
Berkely **S**oftware **D**istribution [1978-1995]

Bl.a. forskning för **ARPAnet**

- Första **UNIX** OS med NCP stöd
- Tidiga med TCP/IP stöd

[3]

Fritt tillgänglig **källkod**



Arpanet Points in the 70s, by Semaforo GMS, 2022, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Arpanet_in_the_1970s.png). CC BY-SA 4.0 International

BSD Descendants

En familj av **BSD** OS

[8]

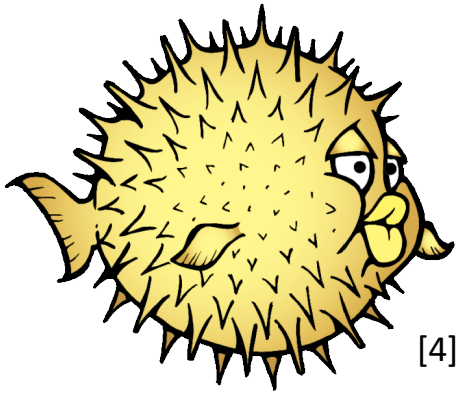
FreeBSD

[1993]

[6]

NetBSD

[1993]



[4]

OpenBSD

[1996]

NEXTSTEP

[1995]

[7]

[5]

Darwin

[2000]

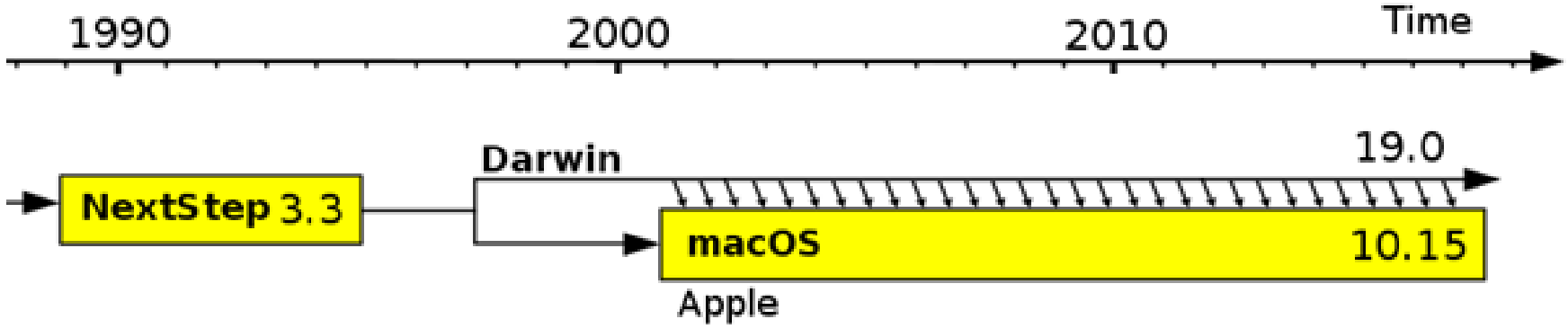
macOS [1999]

NEXTSTEP

[1995] [7]

Darwin [5]

[2000]



Varför BSD?

- Populär server OS… Linux har dock tagit över allt mer
- Baserad på forskning
- Byggt för TCP/IP
- Flera mjukvaror som idag är industristandard

https://people.freebsd.org/~gallatin/talks/euro2021.pdf

Commercial UNIX [1982-1997]

1982: AT&T monopol bryts, **UNIX** börjar säljas

1982 **UNIX System III**

1983 **UNIX System V, SysV**

AT&T UNIX System V license plate, displayed in Japan, by Takuya Oikawa, 2007, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:ATT_UNIX_System_V_License_Plate.jpeg). CC BY-SA 2.0 Generic

Commercial UNIX Derivatives

Flera företag & produkter uppkom runt **UNIX**, b.la.

[1982/1992]

[1982]

[1986]

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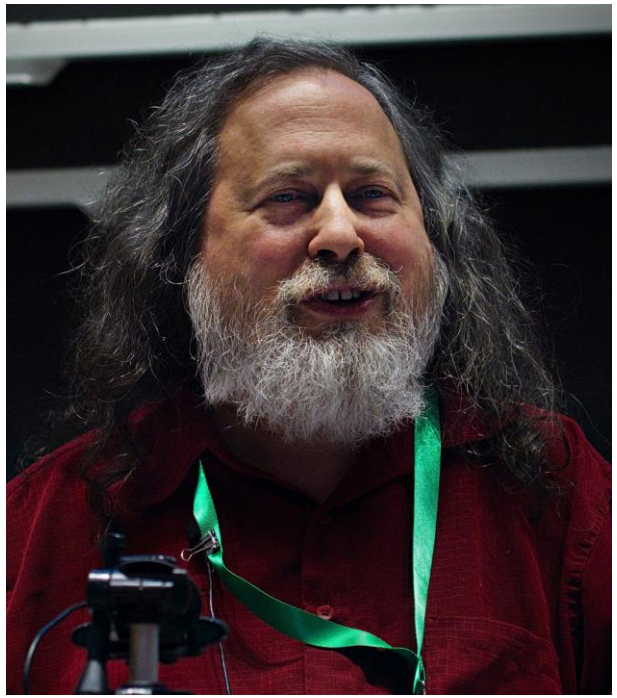
The logo of IBM AIX – Unix operating system from IBM, by IBM, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:IBM_AIX_logo_(2021).svg). Public domain, trademark of HP Inc.

GNU’s **N**ot **U**nix! [1983]

Richard Stallman grundade

- **Free Software Foundation**
- **GNU Project**



Richard Stallman, at LibrePlanet 2019, by Ruben Rodriguez, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=79484097). CC BY 4.0 International

GNU’s **N**ot **U**nix! [1983]

GNU-program

Bash, Coreutils,
Emacs, GRUB,
GCC, make, nano…

GNU-kärna

HURD

Official GNU Head, 2006, by Victor Siame, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Official_gnu.svg). Free Art License

Linux [1991]

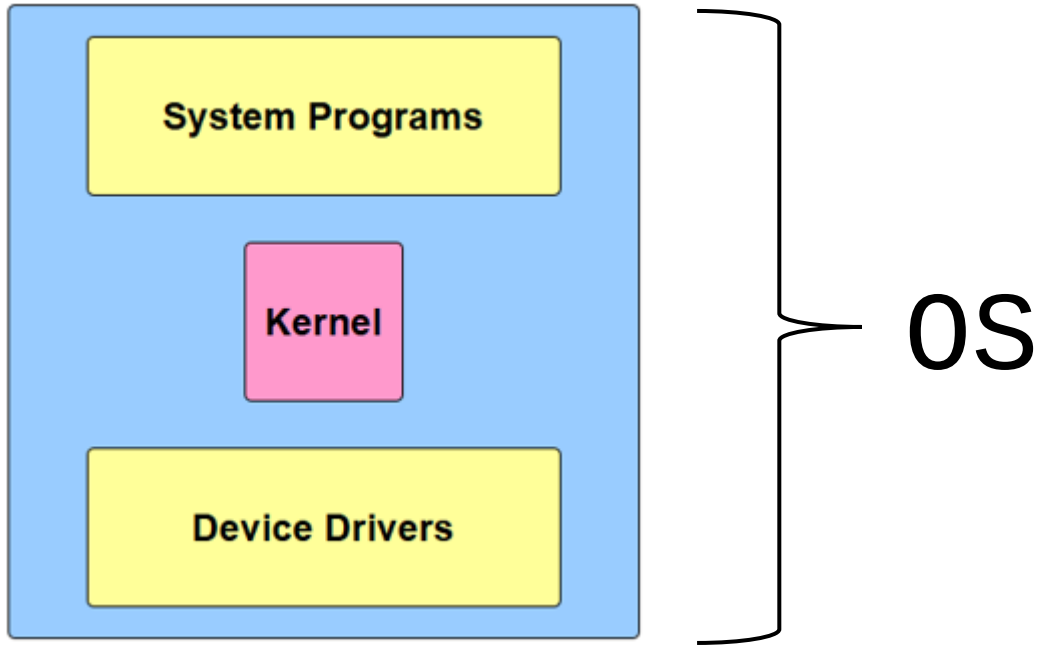
Linus Torvalds

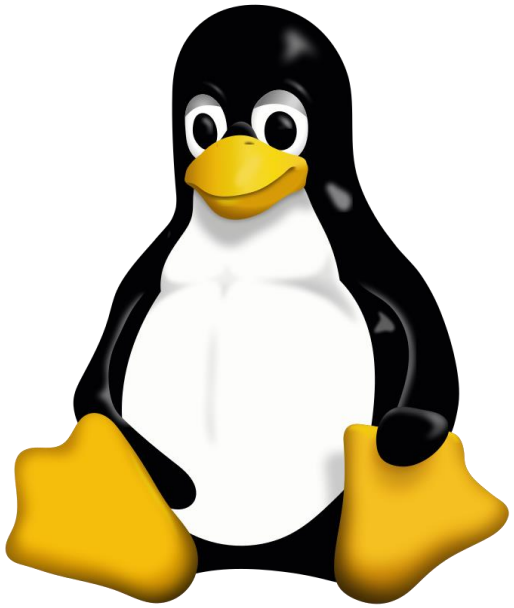
```
Hello everybody out there using minix -

I'm doing a (free) operating system (just a
hobby, won't be big and professional like
gnu) for 386(486) AT clones.
```

A cropped and rotated version of an image of Linus Torvalds speaking at the LinuxCon Europe 2014 in Düsseldorf, by Krd, Von Sprat, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=54706023). CC BY-SA 3.0 Unported

Linux [1991]

UNIX-liknande OS kärna





penguin Tux, the Linux Mascot, by Larry Ewing, Simon Budig, Garrett LeSage, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=753970). CC0 1.0 Universal Public Domain

The UNIX Wars [ca. 1980-2000]

Massor av **UNIX-liknande** OS men ingen interoperabilitet

Original UNIX license plate in the NH style, by Armandops, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Actual_DEC_UNIX_License_Plate_DSC_0317.jpg). CC BY-SA 3.0 Unported

Mt. Xinu Mach 386, Mt Xinu BSD Poster, by Mt Xinu archiver neozeede, Internet Archive (https://archive.org/details/mtxinu-mach386/Mt.Xinu%20BSD%20Poster.png). Fair use/I enlighet med citaträtt

UNIX standardisation

1988 **POSIX**
standard för UNIX-interoperabilitet

1994 **Single UNIX Specification**
certifiering krävs för att få kalla sig **UNIX**

UNIX®
An Open Group Standard

logo for Unix operating systems conforming to the Single UNIX Specification, by Robin Nicholas, Patricia Saunders, The Open Group, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:UNIX_logo.svg). Public domain, trademark of The Open Group

Vad hände med **UNIX**?

finns inget modernt **UNIX** OS…

Ytterst få ***nix** OS är certifierade

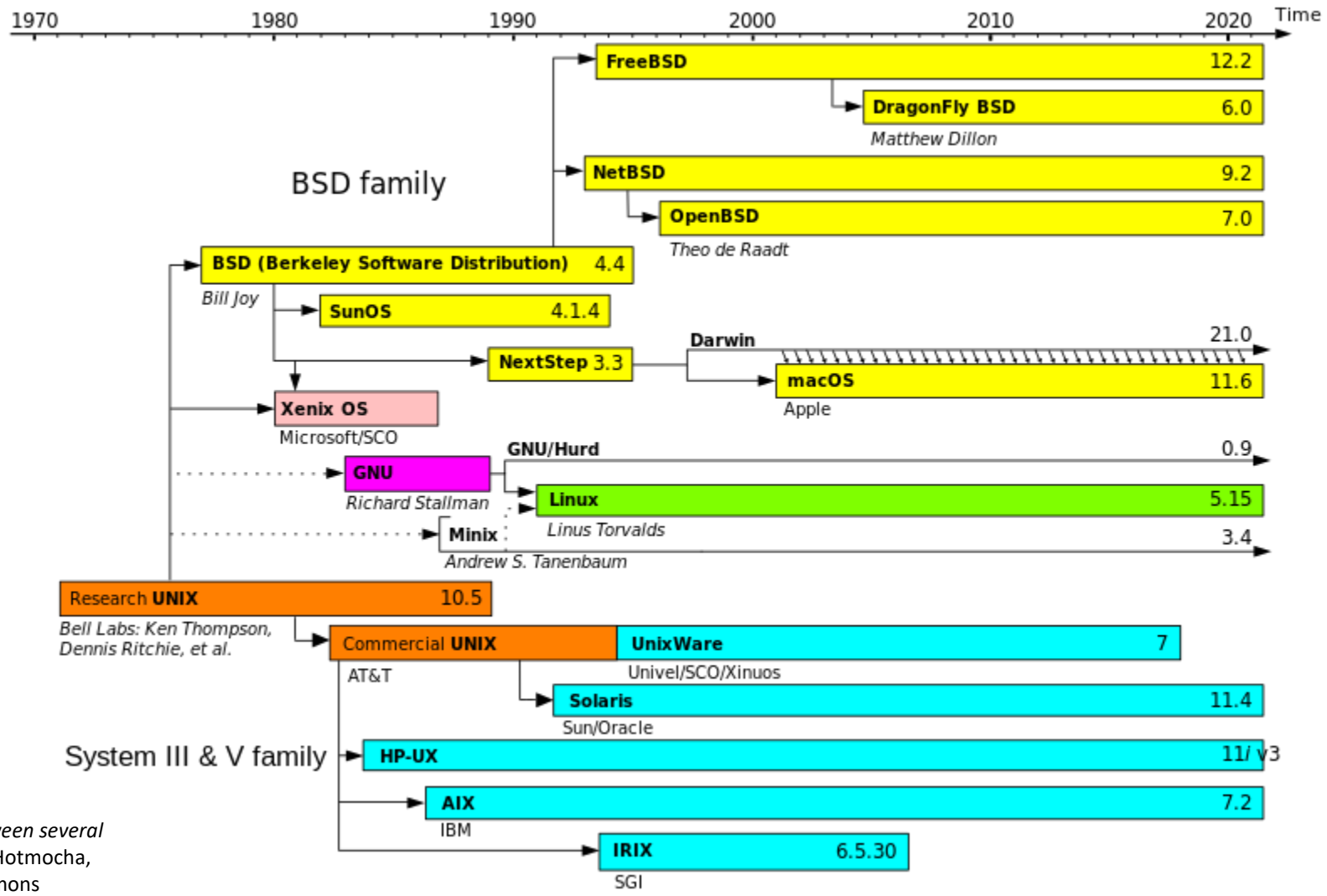
många är dock compliant

Moderna ’**UNIX**’

IBM z/OS & AIX **HP-UX**

UNIX-like

En *nix-familj



A picture showing the relations between several Unix systems, by Guillem, Wereon, Hotmocha, Christoph S., 2009, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Unix_timeline.en.svg). Public domain

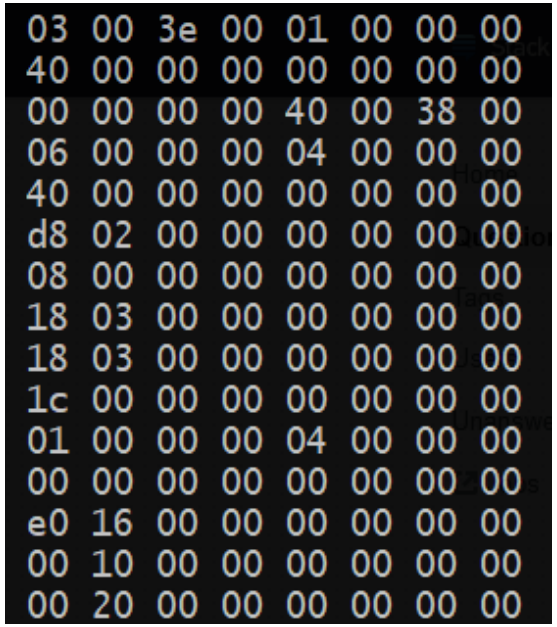
Licenser

Proprietär mjukvara

Oftast:

- Sälj genom en användarlicens
- Ges enbart ut kompilerad
- Tar betalt för mjukvaran

Ex.



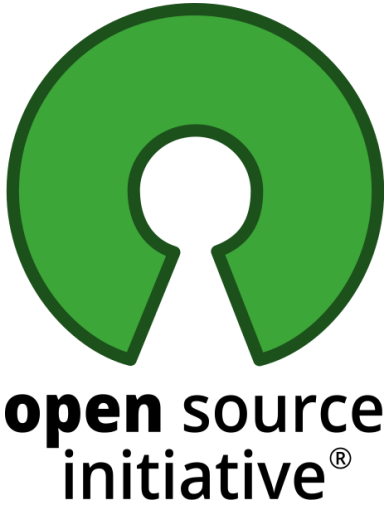
Alltid juridiskt skyddad

- Får inte ändra
  eller återanvända

Open Source

Bland annat

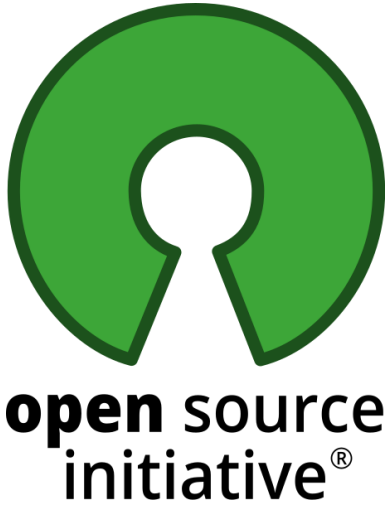
- Apache
- Creative Commons
- CCDL
- BSD
- MIT
- Mozilla



Logo of the Open Source Initiative, "combines the 'O' of open and a keyhole, for unlocking source code, by Colin Viebrock, 2014, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:Open_Source_Initiative.svg). CC BY 2.5 Generic

Open Source

1. Obegränsad vidaredistribution
2. Distribution av källkod och motsvarande kompilerad kod
3. Tillåter modifiering, och distribution av dessa verk under samma licensvillkor



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Free Software

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GNU General Public License

Frihet 0: Friheten att använda programvaran i valfritt syfte

Frihet 1: Friheten att undersöka programmet för att förstå hur det fungerar och använda dessa kunskaper för egna behov

Frihet 2: Friheten att vidaredistribuera kopior för att hjälpa andra

Frihet 3: Friheten att förbättra programmet, anpassa det till egna krav och distribuera förbättringarna så att andra kan dra nytta av modifieringarna

Free, Open, or Proprietary?

Måste inte stämma:

Gratis ≠ **free software** eller **open source**

Kostar ≠ **proprietär mjukvara**

Free software & **open source** är mycket snarlika, mer filosofiska skillnader

’frihet’ vs. samarbete

Free, Open, or Proprietary?

Ex.

Free Software: Linux & GNU

Open Source: majoriteten av BSD-familjen

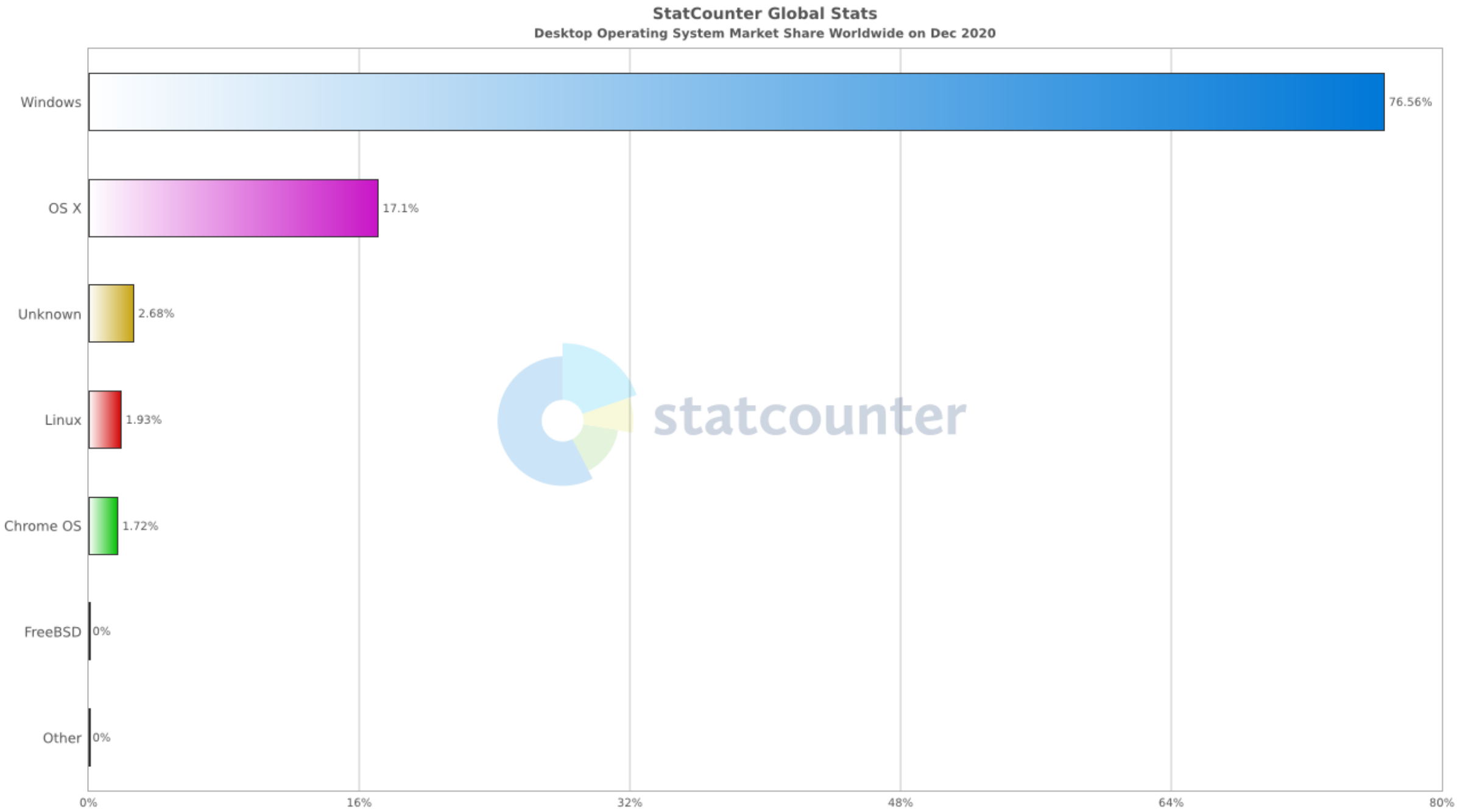
Proprietär: AT&T UNIX, macOS, AIX & HP-UX

Används *nix idag?

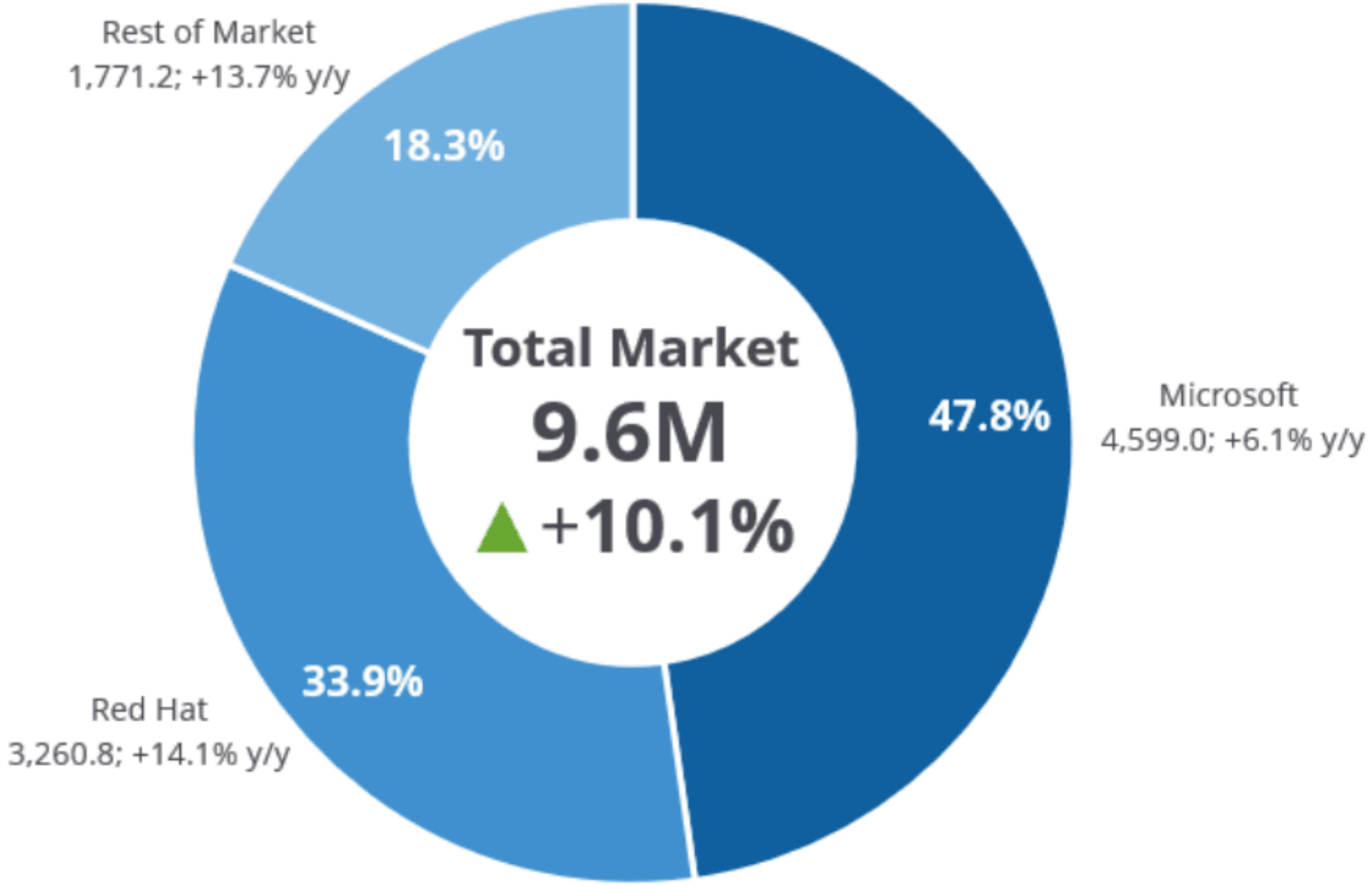
Developers' Primary Operating Systems

All Respondents | Professional Developers

72,590 responses


StatCounter Global Stats
Desktop Operating System Market Share Worldwide on Dec 2020
Windows
76.56%
OS X
17.1%
Unknown
2.68%
Linux
1.93%
Chrome OS
1.72%
FreeBSD
0%
Other
0%
0%
16%
32%
48%
64%
80%
statcounter

Worldwide Server Operating Environments 2018 Share Snapshot



Note: 2018 Share (%), Paid Shipments/Subscriptions (000), and Growth (%)

Source: IDC, 2019

Används *nix idag?

- **Linux** vanligast bland ***nix**
- Du kommer stötta på **Linux**
- Behöver kunna hantera **Linux**

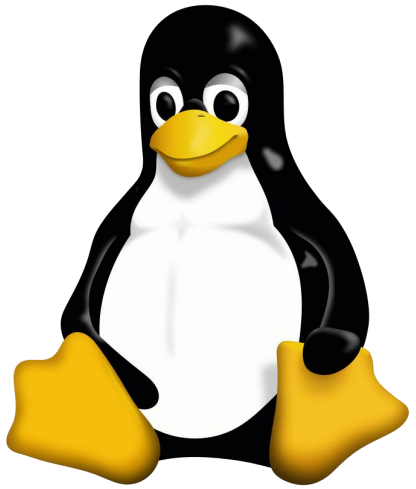
Vad är Linux?
Och hur installerar man det?

Vad är Linux?

UNIX-liknande OS **kärna**

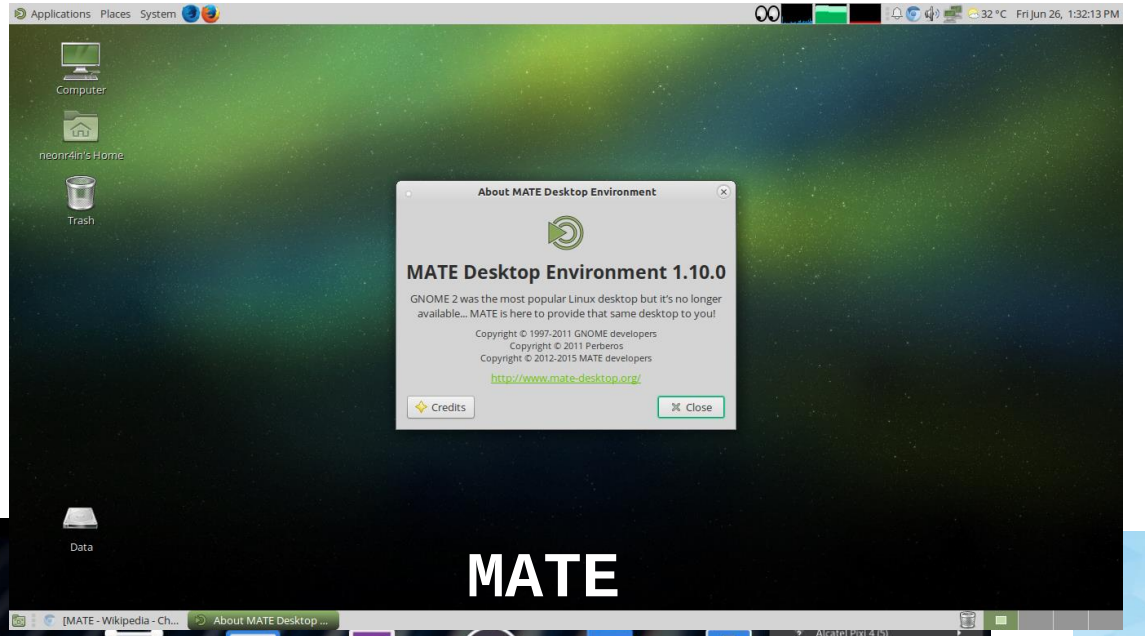
För ett komplett system behövs andra mjukvaror:

- b.la. **GNU Project**
- …eller något liknande

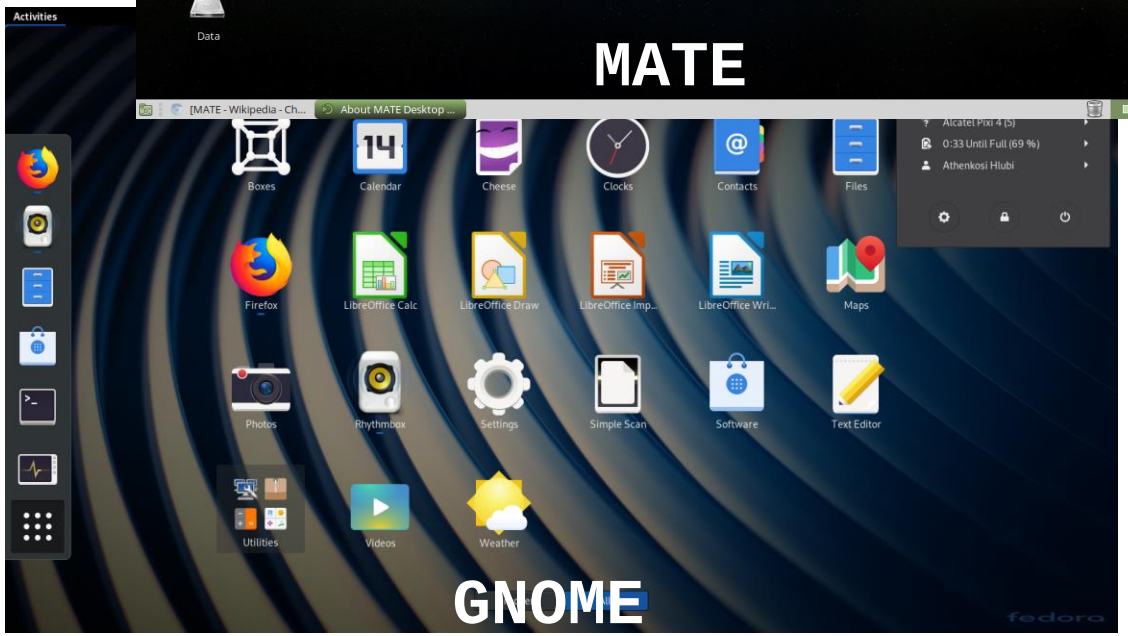


penguin Tux, the Linux Mascot, by Larry Ewing, Simon Budig, Garrett LeSage, Wikimedia Commons (https://commons.wikimedia.org/w/index.php?curid=753970). CC0 1.0 Universal Public Domain

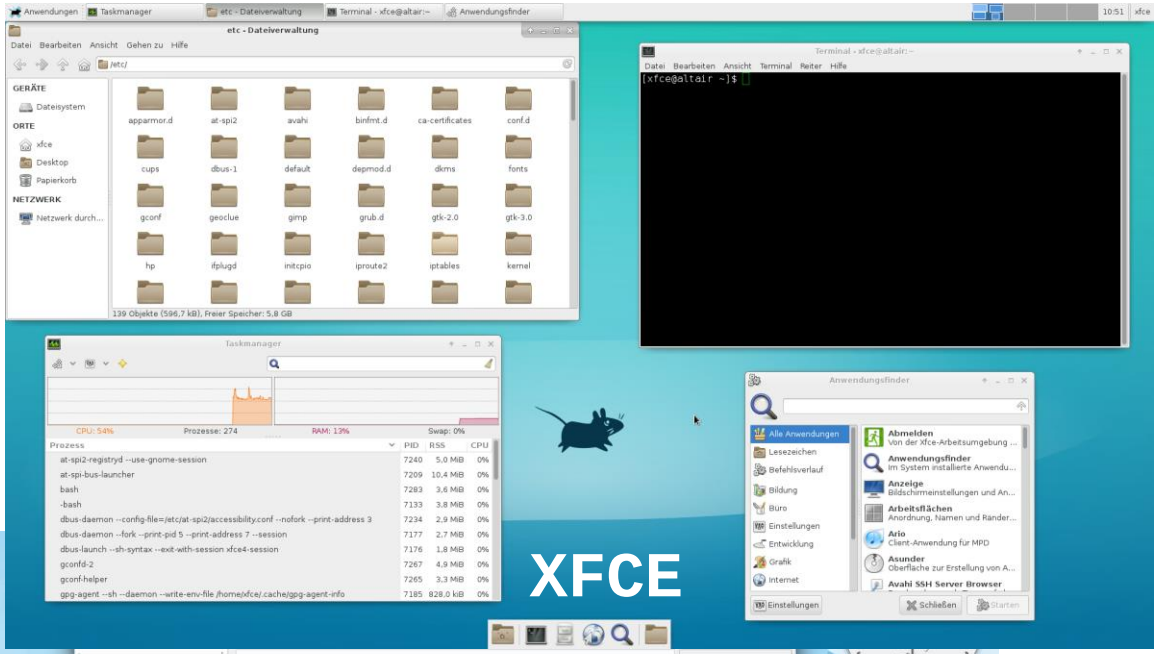
Hur ser Linux ut?



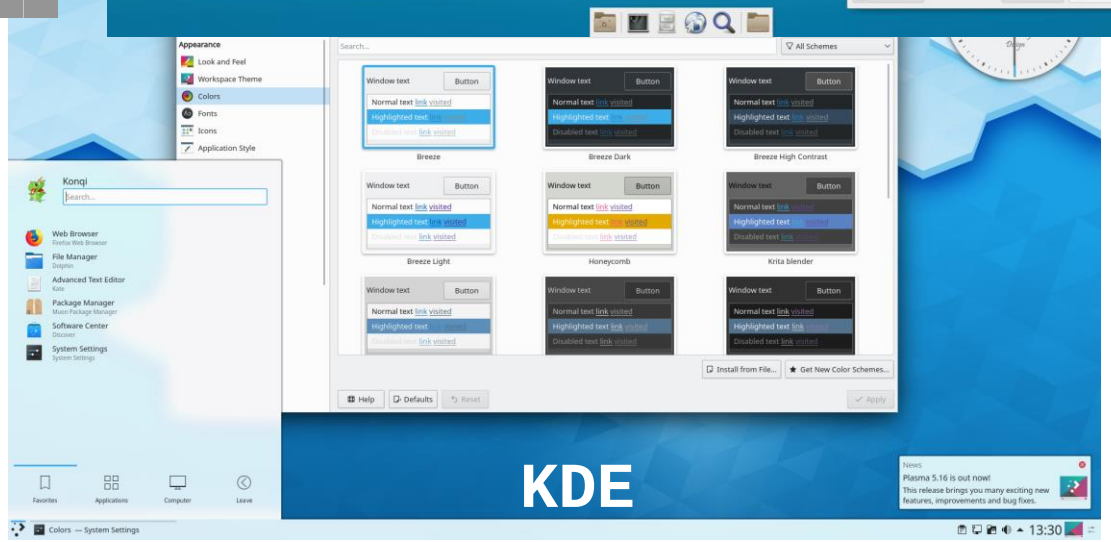
MATE



GNOME

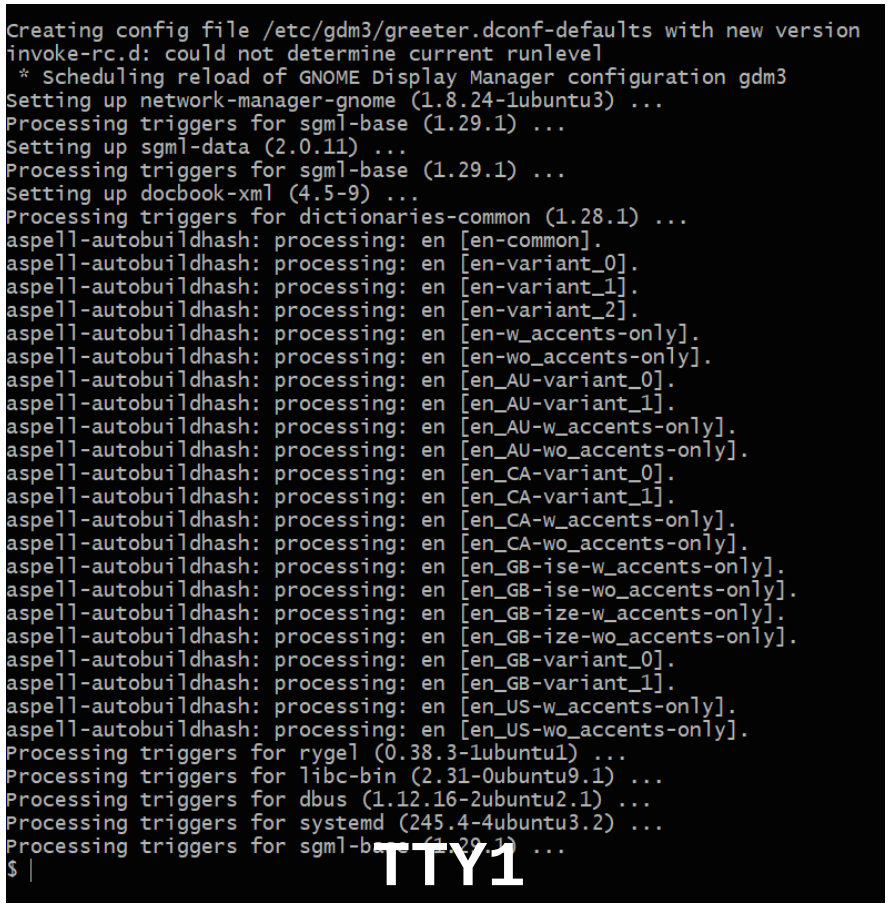


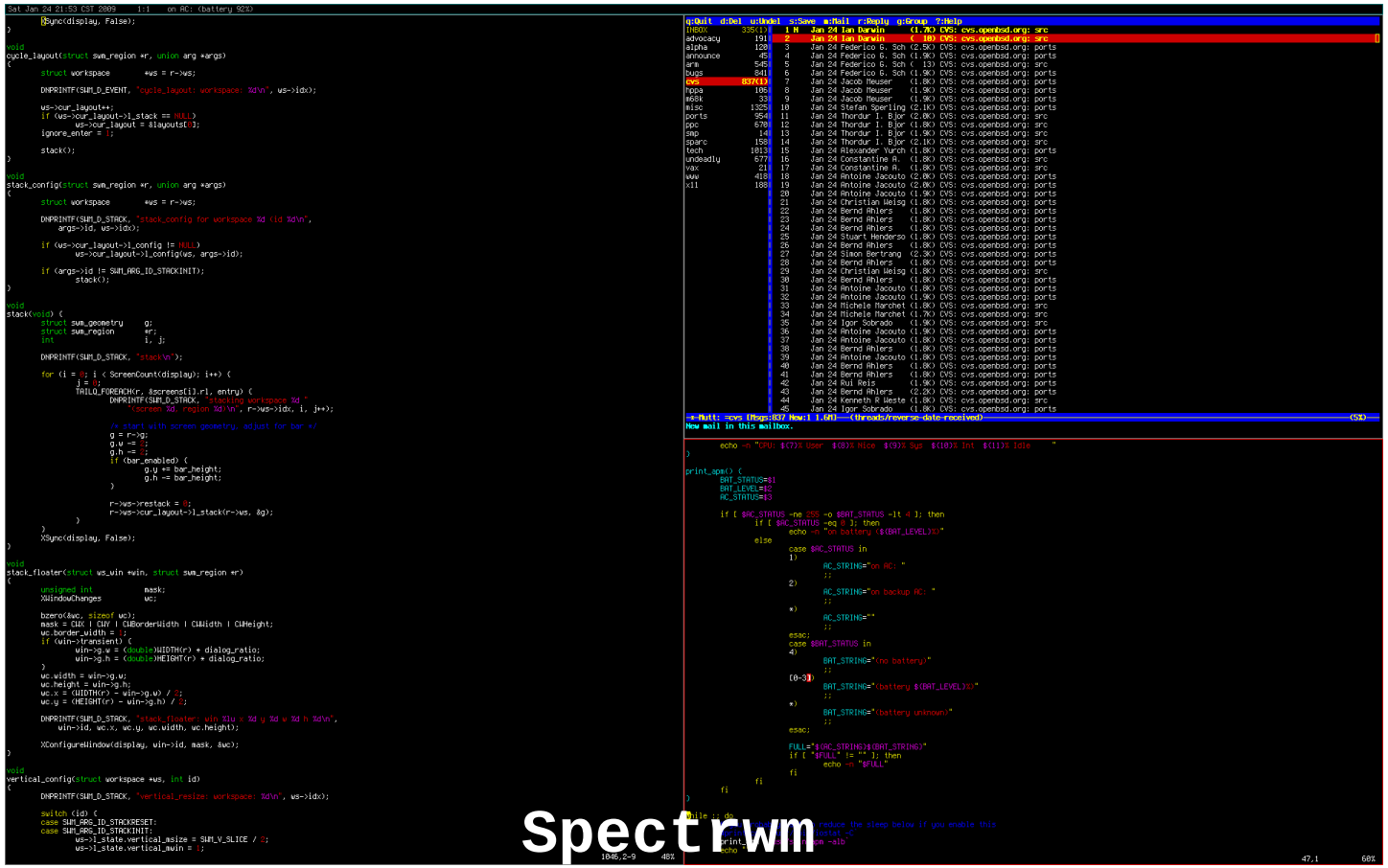
XFCE



KDE

Hur ser Linux ut?





Vad kallas det?

https://en.wikipedia.org/wiki/GNU/Linux_naming_controversy#Opinions

GNU/Linux

Enligt **FSF** & **GNU**

Linux

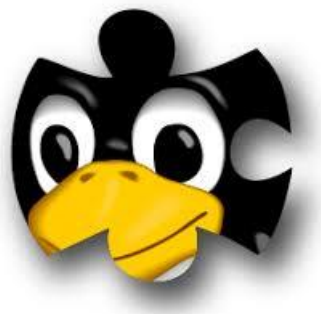
Majoriteten av användare

Tux hugging GNU, by Alsion Upton, Wikimedia Commons (https://commons.wikimedia.org/wiki/File:GNU_Love.png). CC BY 3.0 Unported

Hur installerar man Linux?

Alternativ

1. Ifrån **källkod**
2. Ifrån en **distribution**



Har du använt någon Linux-dist?

Linux Distribution

’distro’

Samling mjukvaror paketerade
i ett komplett **Linux** OS

https://distrowatch.org

Linux Distribution

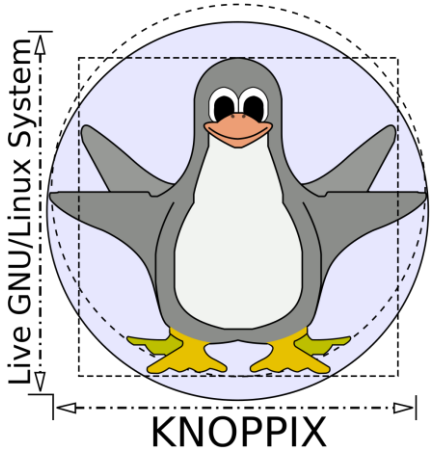
https://upload.wikimedia.org/wikipedia/commons/1/1b/Linux_Distribution_Timeline.svg

Fler distar…

Finns mängder med distributioner…

För **live CD/USB**

körs enbart volatilt i RAM



Vad kommer vi använda?

Linux Distributions Types

Release model

- **Fixed release**
- **Rolling release**

Fixed Release

- Större uppdateringar
- Uppdateras mer sällan, men mer väl testade
- Svårare att uppgradera till en ny OS version

  → Stabilare?

Rolling Release

- Kontinuerliga uppdateringar
- Inga OS versioner
- Senaste versionen av mjukvara

  → Särkare?

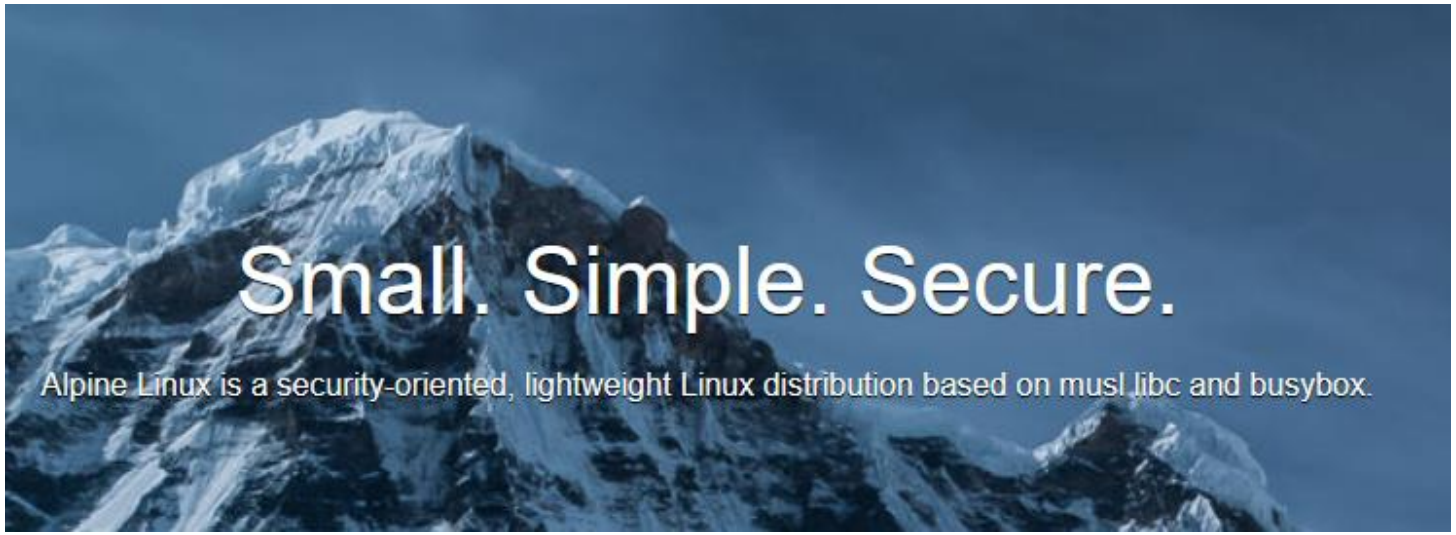
Systemadministration – Linux

Linux för serverbruk

- Distributioner för serverbruk
- Fixed release
- CLI!

Vad kommer vi använda?

- Alpine Linux
- Debian GNU/Linux
- Red Hat Enterprise Linux
- openSUSE Leap



Mycket populär för virtualiserade servrar

Inga GNU-komponenter

debian

Mycket populär distro

Har inspirerat mängder av Debian-baserade distributioner, b.la.

ubuntu

Red Hat
Enterprise Linux

Red Hat ett mjukvaruföretag med b.la. flera distar

För serverbruk: RHEL

Licensbelagt, säljs med support

CentOS

Stöds av Red Hat, baserad på RHEL

100% bug-for-bug compatible with RHEL

https://www.centos.org/centos-linux-eol/

…och nu då?

CentOS Stream

Uppdateringar testas i **CentOS Stream** innan de kommer till **RHEL**

AlmaLinux

Rocky Linux

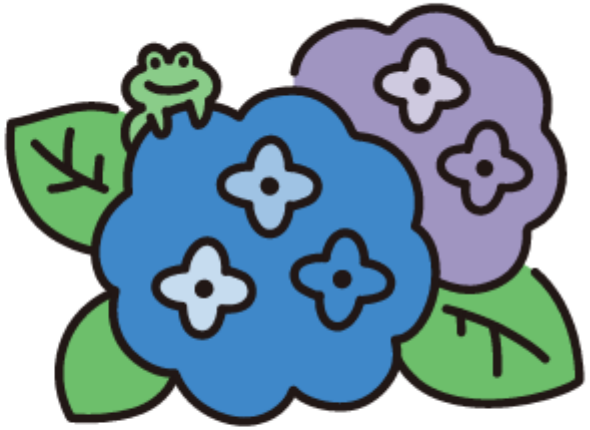
Ska motsvara gamla CentOS men utan Red Hats stöd

kanske snart…

Open Enterprise Linux Association

OpenELA is a trade association of open source Enterprise Linux distribution developers

Read The Announcement



https://openela.org/

SUSE ett mjukvaruföretag med b.la. flera distar

För serverbruk: **SLES**

Licensbelagt, säljs med support

openSUSE Leap motsvara vad gamla **CentOS** var för **RHEL**

Till nästa gång

- Läs kurslitteraturen
- Gå med i labbgrupp
- tisdag: föreläsning, Linux CLI

Tack för idag

Bildreferenser

[1] "*Photo of Unix creators Ken Thompson (left) and Dennis Ritchie (right) from the latest edition of the Jargon File*" [Online image]. Wikimedia Commons. Available: https://commons.wikimedia.org/wiki/File:Ken_Thompson_and_Dennis_Ritchie--1973.jpg [Accessed: January 10, 2024]

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